Status	Started	Completed
Finished	Wednesday, 11 November 2026, 1:11 PM	Wednesday, 11 November 2026, 1:13 PM

Duration	Grade
1 min 11 secs	**20.00** out of 20.00 (**100**%)

Question 1

Flag question

Correct Mark 1.00 out of 1.00

A customer support conversation runs 35 turns. The agent has been compressing history every 10 turns. In the final turns, the agent seems to have forgotten key details from turns 5-15 where important case context was established. Progressive summarisation condensed these turns into vague abstractions. What architectural pattern specifically preserves case-critical details across compressions?

a. Avoid using compression in support conversations — always maintain full history.

b. Extract case-critical facts — account identifiers, amounts, product names, dates, commitments made — into a persistent 'case facts' block that exists outside the compressed conversation history. This block is never subject to compression and is included at the top of every prompt, ensuring critical specifics survive across multiple compression cycles.

c. Add a verification step at the start of each turn asking the customer to confirm key details.

d. Use a longer summarisation interval to reduce the number of compression events.

The correct answer is: Extract case-critical facts — account identifiers, amounts, product names, dates, commitments made — into a persistent 'case facts' block that exists outside the compressed conversation history. This block is never subject to compression and is included at the top of every prompt, ensuring critical specifics survive across multiple compression cycles.

Question 2

Flag question

Correct Mark 1.00 out of 1.00

Your pipeline routes 58% of documents to human review because any field with confidence below 0.88 triggers review. Manual review shows 93% of reviewed documents are correctly extracted. Reviewers are overwhelmed. What does this indicate and how do you fix it?

a. The 0.88 threshold was set without calibrating against actual error rates. If 93% of reviewed documents are correct at confidence 0.87, then 0.88 is routing far too many documents for review. Calibrate using a labeled validation set: find the confidence level where the actual error rate becomes unacceptable, and set the threshold there rather than using an arbitrary round number.

b. The 0.88 threshold is appropriate — high-accuracy systems require stringent review.

c. Lower the threshold to 0.75 to reduce review volume.

d. Replace confidence-based routing with value-based routing.

The correct answer is: The 0.88 threshold was set without calibrating against actual error rates. If 93% of reviewed documents are correct at confidence 0.87, then 0.88 is routing far too many documents for review. Calibrate using a labeled validation set: find the confidence level where the actual error rate becomes unacceptable, and set the threshold there rather than using an arbitrary round number.

Question 3

Flag question

Correct Mark 1.00 out of 1.00

Your agent handles customer account changes. In turn 3, the customer confirms their new email as 'john.doe@company.com'. In turn 25, while processing an account update, the agent uses 'j.doe@company.com' — a subtly different address that was never stated. Where did this error come from and how is it prevented?

a. The customer made a typo in their original email submission.

b. The agent tool retrieved an outdated email from the database.

c. The model hallucinated a plausible variation of the email address. Exact values like email addresses, account numbers, and identifiers must be stored verbatim in an immutable facts block at session start and referenced directly from that block when needed for transactions — never reconstructed from memory or context.

d. The email was compressed during conversation history summarisation.

The correct answer is: The model hallucinated a plausible variation of the email address. Exact values like email addresses, account numbers, and identifiers must be stored verbatim in an immutable facts block at session start and referenced directly from that block when needed for transactions — never reconstructed from memory or context.

Question 4

Flag question

Correct Mark 1.00 out of 1.00

Your research agent session correctly identifies a crucial insight about market structure in hour 2 of an 8-hour session. When the developer asks about this insight in hour 7, the agent gives an inconsistent answer. /compact was run in hour 4. What does this reveal about /compact and key findings?

a. /compact deleted the insight from the session — do not use /compact during investigations.
b. /compact compresses verbose conversation history but may lose specific details in the process. The insight from hour 2 was compressed into a general summary that lost the specific detail. Key insights must be explicitly written to a scratchpad or notes file before running /compact — external persistence is necessary for findings that must survive context compression.
c. The agent is correct in hour 7 — the hour 2 insight was probably wrong.
d. /compact should be run more frequently to keep context fresh.

The correct answer is: /compact compresses verbose conversation history but may lose specific details in the process. The insight from hour 2 was compressed into a general summary that lost the specific detail. Key insights must be explicitly written to a scratchpad or notes file before running /compact — external persistence is necessary for findings that must survive context compression.

Question 5

Flag question

Correct Mark 1.00 out of 1.00

Your extraction pipeline processes documents in multiple languages. Your aggregate accuracy is 92% but you have tested only on English and Spanish documents. German, French, and Japanese documents are now being processed. What should happen before these new languages are added to automated processing?

a. Use the same confidence threshold for all languages — the model calibrates automatically.
b. The 92% overall accuracy applies to all languages — proceed with automation.
c. Add the new languages to the extraction prompt and monitor accuracy for 30 days.
d. Validate extraction accuracy specifically on representative samples from each new language (German, French, Japanese) before automating. Languages with different character sets, grammar structures, and number formatting conventions may perform significantly differently from the tested languages. Per-language accuracy validation is required before each language is included in automated processing.

The correct answer is: Validate extraction accuracy specifically on representative samples from each new language (German, French, Japanese) before automating. Languages with different character sets, grammar structures, and number formatting conventions may perform significantly differently from the tested languages. Per-language accuracy validation is required before each language is included in automated processing.

Question 6

Flag question

Correct Mark 1.00 out of 1.00

Your agent processes 500 invoices per hour. A critical payment validation service goes down for 45 minutes. What is the correct operational pattern during the outage?

a. Reduce the pipeline to 10% throughput until the validation service recovers.
b. Continue processing invoices that do not require payment validation. Tag invoices requiring validation with 'payment_validation_pending' status and queue them in a separate queue. When the service recovers, process the queued invoices. Structured partial operation maintains throughput for the majority while tracking the minority requiring the unavailable service. ✓
c. Process all invoices and skip validation for all 45 minutes to maintain throughput.
d. Stop the entire invoice pipeline until the payment validation service recovers.

The correct answer is: Continue processing invoices that do not require payment validation. Tag invoices requiring validation with 'payment_validation_pending' status and queue them in a separate queue. When the service recovers, process the queued invoices. Structured partial operation maintains throughput for the majority while tracking the minority requiring the unavailable service.

Question 7

Flag question

Correct Mark 1.00 out of 1.00

Your multi-source research pipeline produces a synthesis report claiming 'AI revenue grew 40% in 2024.' After publication, a client notices the figure conflicts with industry reports they have seen. The coordinator cannot identify which subagent or document produced this figure. What retroactive fix reveals the source?

a. There is no way to trace the figure retroactively once it has been synthesised.
b. This is only traceable if claim-source mappings were captured at subagent output time. Add a requirement for future runs: all subagents must output structured claim-source mappings (claim text + source + date + relevant excerpt) that the synthesis agent preserves. Retroactive tracing of already-merged claims is generally not possible — provenance must be built in upstream. ✓
c. Search all source documents for the 40% figure as a post-synthesis step.
d. Query each subagent individually asking whether they produced the 40% figure.

The correct answer is: This is only traceable if claim-source mappings were captured at subagent output time. Add a requirement for future runs: all subagents must output structured claim-source mappings (claim text + source + date + relevant excerpt) that the synthesis agent preserves. Retroactive tracing of already-merged claims is generally not possible — provenance must be built in upstream.

Question 8

Flag question

Correct Mark 1.00 out of 1.00

Your pipeline collects confidence scores from a model and uses a fixed threshold of 0.90 to route to human review. A new model version is deployed. Previously, scores of 0.90 corresponded to 95% accuracy. After the model update, scores of 0.90 correspond to 88% accuracy for the same types of documents. What must happen when model versions change?

a. The 0.90 threshold remains valid — confidence scores are calibrated universally across model versions.
b. The pipeline does not need updating — model versions do not affect calibration.
c. Increase the threshold to 0.95 after model updates as a precautionary measure.
d. Re-calibrate the routing threshold after every model version change. Confidence scores are not standardised across models — a 0.90 from Model V1 and a 0.90 from Model V2 may represent different actual accuracy levels. Recalibrate using a labeled validation set after any model update to maintain intended accuracy levels. ✓

The correct answer is: Re-calibrate the routing threshold after every model version change. Confidence scores are not standardised across models — a 0.90 from Model V1 and a 0.90 from Model V2 may represent different actual accuracy levels. Recalibrate using a labeled validation set after any model update to maintain intended accuracy levels.

Question 9

Flag question

Correct Mark 1.00 out of 1.00

Your agent workflow includes a step that sends automated SMS alerts to 10,000 customers when a product is back in stock. During a test run, a loop condition causes the alert to be sent 8 times to each customer. What safeguards prevent this from reaching production?

a. Add a catch for the loop condition that caused the over-sending.
b. Add max_iterations: 1 on the SMS send step.
c. Implement idempotency at both the tool and workflow levels: an idempotency_key per customer per restock event prevents duplicate sends even if the tool is called multiple times; a loop-detection guard tracks tool call fingerprints and breaks when the same tool is called with identical parameters more than once per session. ✓
d. Route all SMS sends through a human approval step before execution.

The correct answer is: Implement idempotency at both the tool and workflow levels: an idempotency_key per customer per restock event prevents duplicate sends even if the tool is called multiple times; a loop-detection guard tracks tool call fingerprints and breaks when the same tool is called with identical parameters more than once per session.

Question 10

Flag question

Correct Mark 1.00 out of 1.00

Your agent for contract analysis processes a 120-page enterprise agreement. The contract is loaded into context at the start of the session. After the initial loading and 20 turns of Q&A about the contract, the agent starts misquoting specific clause numbers and amounts. What is the cause?

a. Increase max_tokens to accommodate both the 120-page contract and 20 turns of history.
b. As the conversation history grows to 20 turns, the 120-page contract content gets pushed toward the middle of the context window where the 'lost in the middle' effect causes reduced attention to specific details. Mitigate by: placing the most frequently referenced clauses near the end of the context, using a structured facts block for critical clause numbers and amounts, or using a retrieval-based approach for specific clause lookups. ✓
c. The model cannot maintain attention on specific details from a 120-page document across 20 turns of interaction.
d. The contract must be re-loaded every 10 turns to maintain attention on its content.

The correct answer is: As the conversation history grows to 20 turns, the 120-page contract content gets pushed toward the middle of the context window where the 'lost in the middle' effect causes reduced attention to specific details. Mitigate by: placing the most frequently referenced clauses near the end of the context, using a structured facts block for critical clause numbers and amounts, or using a retrieval-based approach for specific clause lookups.

Question 11

Flag question

Correct Mark 1.00 out of 1.00

Your coordinator spawns subagents but one subagent (document_analysis) consistently returns less valuable findings than expected. Investigation shows it receives a Task prompt that says: 'Analyse the documents and return your findings.' The web search subagent receives a detailed Task prompt specifying documents to analyse, quality criteria, output format, and expected depth. What explains the performance difference?

a. The document_analysis subagent's model is less capable than the web search subagent's model.
b. The document_analysis subagent is processing more complex documents.
c. The document_analysis subagent needs better tools.
d. Subagent performance is directly proportional to Task prompt quality. The vague document_analysis prompt produces vague findings; the detailed web search prompt produces specific, useful findings. Every subagent Task prompt should specify: the exact documents or data to process, the specific questions to answer, the output format required, and quality criteria. Vague delegation produces vague results. ✓

The correct answer is: Subagent performance is directly proportional to Task prompt quality. The vague document_analysis prompt produces vague findings; the detailed web search prompt produces specific, useful findings. Every subagent Task prompt should specify: the exact documents or data to process, the specific questions to answer, the output format required, and quality criteria. Vague delegation produces vague results.

Question 12

Flag question

Correct Mark 1.00 out of 1.00

During a 5-hour codebase exploration session, /compact is run at hour 3. After compaction, the agent gives inconsistent answers about a critical database model identified in hour 1. What caused this and what should have been done before /compact?

a. /compact compresses verbose conversation history which may lose specific details like exact model names and field definitions. Before running /compact, the agent should have written the critical database model details — table names, fields, relationships — to a scratchpad file. Explicit external persistence preserves specifics through compaction.

b. /compact is not suitable for sessions involving database models.

c. The model has a memory limitation — /compact does not preserve any prior findings.

d. The database model findings should have been committed to the repository before /compact.

The correct answer is: /compact compresses verbose conversation history which may lose specific details like exact model names and field definitions. Before running /compact, the agent should have written the critical database model details — table names, fields, relationships — to a scratchpad file. Explicit external persistence preserves specifics through compaction.

Question 13

Flag question

Correct Mark 1.00 out of 1.00

Your pipeline processes financial reports where numbers are sometimes stated in thousands ('$1,500' meaning $1,500,000) and sometimes in full ('$1,500,000'). After automated extraction, some downstream calculations are off by a factor of 1,000. What extraction design prevents this scale error?

a. Always assume all numbers are in full dollars.

b. Add a scale_warning field that flags documents using thousands notation.

c. Design the extraction to identify and normalise the document's numerical scale convention: instruct the model to detect scale context (e.g. '(in thousands)' headers, table footnotes) and normalise all extracted values to full absolute dollars. Include few-shot examples showing scale detection and normalisation for both conventions.

d. Accept scale errors and fix them in downstream application code.

The correct answer is: Design the extraction to identify and normalise the document's numerical scale convention: instruct the model to detect scale context (e.g. '(in thousands)' headers, table footnotes) and normalise all extracted values to full absolute dollars. Include few-shot examples showing scale detection and normalisation for both conventions.

Question 14

Flag question

Correct Mark 1.00 out of 1.00

Your pipeline processes 1,500 loan applications per day. A third-party credit check API goes down for 2 hours during peak hours, affecting 500 applications. What is the correct handling during the outage?

a. Redirect all 500 applications to a backup credit check provider immediately.
b. Process the non-credit-check stages of all 500 applications and store them with status 'credit_check_pending'. When the API recovers, automatically trigger credit checks for all pending applications. This maximises the work completed during the outage and provides a clear recovery path. ✓
c. Stop processing all applications until the credit check API recovers.
d. Process all 500 applications without credit checks to maintain throughput, flagging them for manual review later.

The correct answer is: Process the non-credit-check stages of all 500 applications and store them with status 'credit_check_pending'. When the API recovers, automatically trigger credit checks for all pending applications. This maximises the work completed during the outage and provides a clear recovery path.

Question 15

Flag question

Correct Mark 1.00 out of 1.00

Your agent session is used to investigate a complex authentication bug. During the investigation, the agent identifies 5 critical issues with the session management implementation. The developer ends the session to implement fixes. The next day, they need the agent to verify the fixes. What is the best approach for the follow-up session?

a. Resume the prior session using --resume to maintain full continuity.
b. Start a fresh session and inject: (1) a structured summary of the 5 identified issues, (2) the specific files that were changed in implementing the fixes. The prior session's tool results are stale since files were modified. A targeted fresh session with relevant context is more reliable than resuming with 24-hour-old file contents. ✓
c. Fork the prior session to create an independent verification branch.
d. Ask the agent to re-investigate from scratch to avoid prior session bias.

The correct answer is: Start a fresh session and inject: (1) a structured summary of the 5 identified issues, (2) the specific files that were changed in implementing the fixes. The prior session's tool results are stale since files were modified. A targeted fresh session with relevant context is more reliable than resuming with 24-hour-old file contents.

Question 16

Flag question

Correct Mark 1.00 out of 1.00

Your pipeline processes supplier invoices from 200 different suppliers. Overall accuracy is 95%. An audit reveals that 8 specific suppliers (representing 4% of volume) have accuracy below 60%. The other 192 suppliers average 97%. What operational decision should be made based on this segmented data?

a. Continue automated processing for all 200 suppliers — 95% overall is acceptable.
b. Remove the 8 problematic suppliers from the automated pipeline permanently.
c. Add human review for all 200 suppliers until the 8 problematic ones are fixed.
d. Automate processing for the 192 high-accuracy suppliers. Route invoices from the 8 low-accuracy suppliers to manual review or to a specialised extraction model trained on their formats. Segmented accuracy data enables targeted operational decisions rather than either accepting poor performance or over-engineering the high-accuracy majority. ✓

The correct answer is: Automate processing for the 192 high-accuracy suppliers. Route invoices from the 8 low-accuracy suppliers to manual review or to a specialised extraction model trained on their formats. Segmented accuracy data enables targeted operational decisions rather than either accepting poor performance or over-engineering the high-accuracy majority.

Question 17

Flag question

Correct Mark 1.00 out of 1.00

Your research report states: 'AI adoption in manufacturing increased 34% year-over-year.' The coordinator cannot identify which subagent or document produced this figure, or whether it refers to 2022-2023 or 2023-2024. What two structural requirements prevent both problems?

a. Require post-synthesis fact-checking to verify all quantitative claims.
b. Require subagents to output claim-source mappings (claim text + source name + URL + publication date) and require temporal metadata for all statistical claims (data collection period or publication date). Both requirements must be preserved through synthesis. This enables attribution AND temporal context for every statistic. ✓
c. Require the synthesis agent to include footnotes for all statistics.
d. Have the coordinator verify every statistic by querying each subagent about its contributions.

The correct answer is: Require subagents to output claim-source mappings (claim text + source name + URL + publication date) and require temporal metadata for all statistical claims (data collection period or publication date). Both requirements must be preserved through synthesis. This enables attribution AND temporal context for every statistic.

Question 18

Flag question

Correct Mark 1.00 out of 1.00

Your agent processes customer data. A session handling Customer A's complex case is reused for Customer B's session by appending B's messages to the existing history. By turn 15, the agent references Customer A's account details when responding to Customer B. What is the fix?

a. Initiate a fresh session for each customer — never reuse conversation history across different customers. Cross-customer context reuse creates data leakage and incorrect responses. Each session must begin with a clean context initialised only with the current customer's relevant information.

b. Compress Customer A's conversation before starting Customer B's session.

c. Add a system prompt instruction: 'Ignore all prior customer information when a new customer message arrives.'

d. Use /compact to remove all traces of Customer A before Customer B's session begins.

The correct answer is: Initiate a fresh session for each customer — never reuse conversation history across different customers. Cross-customer context reuse creates data leakage and incorrect responses. Each session must begin with a clean context initialised only with the current customer's relevant information.

Question 19

Flag question

Correct Mark 1.00 out of 1.00

Your pipeline processes 10,000 documents per day with a 97% accuracy target. Individual document failures need a recovery mechanism. What is the complete reliability pattern for achieving 97% accuracy at scale?

a. Implement a validation-retry loop: (1) validate each extraction against the JSON schema and business rules, (2) on failure, retry with specific error feedback (field name, violation, expected format) injected into the retry prompt — up to 3 retries, (3) after 3 failures, escalate to human review queue. Track escalation patterns to identify systemic issues for prompt improvement.

b. Run each document through the extraction pipeline twice and flag any differences.

c. Add human review for all 30,000 documents flagged at 97% accuracy.

d. Accept the natural failure rate — at scale, some errors are inevitable.

The correct answer is: Implement a validation-retry loop: (1) validate each extraction against the JSON schema and business rules, (2) on failure, retry with specific error feedback (field name, violation, expected format) injected into the retry prompt — up to 3 retries, (3) after 3 failures, escalate to human review queue. Track escalation patterns to identify systemic issues for prompt improvement.

Question 20

Flag question

Correct Mark 1.00 out of 1.00

Your pipeline extracts structured data from 5 different document templates used by different departments. Templates A, B, and C consistently achieve 96%+ accuracy. Templates D and E consistently achieve 65-70% accuracy. What operational decision is correct based on this segmented data?

a. Apply human review to all 5 templates until all achieve 96%+.
b. Remove templates D and E from the automated system permanently.
c. Accept the mixed accuracy — the aggregate figure is acceptable.
d. Automate processing for templates A, B, and C where accuracy is high. Route templates D and E to human review or to a targeted improvement effort (additional few-shot examples specific to those templates, investigation of systematic errors). Segmented data enables proportionate responses rather than uniform policies that either over-review good performance or under-review poor performance. ✓

The correct answer is: Automate processing for templates A, B, and C where accuracy is high. Route templates D and E to human review or to a targeted improvement effort (additional few-shot examples specific to those templates, investigation of systematic errors). Segmented data enables proportionate responses rather than uniform policies that either over-review good performance or under-review poor performance.

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